2201651-Experiment 9 chi square testing

1. When do we go for Chi-Square test?
We use the Chi-Square test when we want to examine the relationship between categorical variables. It helps determine whether the observed frequency distribution differs significantly from the expected distribution under the null hypothesis.
Commonly used for:

- Testing goodness of fit
- Testing independence between variables in a contingency table

2. Do we use Chi-Square test for independence? Explain How.
Yes, the Chi-Square Test for Independence checks whether two categorical variables are independent of each other.

- We create a contingency table showing the frequency of occurrences for combinations of the variables.
- The test compares the observed frequencies with the expected frequencies (assuming independence).
- A significant p-value (typically < 0.05) suggests that the variables are not independent, i.e., they are related.

3. Explain role of Critical value and p-value in Chi-Square test.

- Critical value is the threshold value from the Chi-Square distribution table based on a chosen significance level (α) and degrees of freedom.
- If the Chi-Square statistic > critical value, we reject the null hypothesis.
- p-value is the probability of getting a test statistic as extreme as the one calculated, assuming the null hypothesis is true.
- If p-value < α (e.g., 0.05), we reject the null hypothesis.

Both are used to decide whether to reject the null hypothesis.

4. Can we use Chi-Square test to show existence of correlation between 2 features in a dataset? How will you interpret test result to indicate correlation presence?
The Chi-Square Test does not measure correlation strength like Pearson's r, but it does show whether a relationship exists between two categorical features.

- If the p-value < 0.05, we conclude that the two variables are dependent, meaning there is a relationship.
- If the p-value > 0.05, we conclude that the two variables are independent, i.e., no relationship. So, the Chi-Square test helps to detect the presence of association, not the strength or direction.

5. What is degrees of freedom and how to calculate it while performing Chi-Square test?
Degrees of freedom (df) represent the number of independent values that can vary in a calculation.

- For a Chi-Square test of independence using a contingency table:
  df = (number of rows - 1) × (number of columns - 1)
  It's used to determine the critical value from the Chi-Square distribution table.